Model Memory vs Retrieval

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected retrieval and routing record.

Explain the index, trace stale evidence, audit a shortcut and design a revealing check.

Keep one habit: Keep source IDs, versions, access conditions and unresolved facts together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Verify Numbers and Dates. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Hallucination Failure Types. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Visual Composition. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Verify Numbers and Dates

Hallucination Failure Types

Visual Composition

W1 · Explain retrieval and indexing

> Separate learned patterns, searchable representation and source passage.

Your evidence record

- Room-index practice: index v1 returns N1 assigning room A. Supplied corpus v2 supersedes N1 with N2 assigning room B. A stale returned record cannot establish current approval. Refresh against the authorised corpus or report unresolved freshness. Separate library case: approved current LIB-v2 says Wednesday closes16:00; older generated output says15:00. If the notice is unavailable, current time is unverified.

- Reference for this exercise: open this week's student lesson, "Model Memory vs Retrieval", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the stale room index

> Retain corpus and index versions in the answer.

Your evidence record

- Room-index practice: index v1 returns N1 assigning room A. Supplied corpus v2 supersedes N1 with N2 assigning room B. A stale returned record cannot establish current approval. Refresh against the authorised corpus or report unresolved freshness. Separate library case: approved current LIB-v2 says Wednesday closes16:00; older generated output says15:00. If the notice is unavailable, current time is unverified.

- **Index** v1 contains notice N1 for room A. **Corpus** v2 supersedes it with N2 for room B, but the query still returns N1. What can be reported? Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Audit a retrieval guarantee

> Similarity does not establish authorisation or current applicability.

Your evidence record

- Room-index practice: index v1 returns N1 assigning room A. Supplied corpus v2 supersedes N1 with N2 assigning room B. A stale returned record cannot establish current approval. Refresh against the authorised corpus or report unresolved freshness. Separate library case: approved current LIB-v2 says Wednesday closes16:00; older generated output says15:00. If the notice is unavailable, current time is unverified.

- A report omits this qualification: High **retrieval** similarity does not override permissions or make an obsolete source current. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Design a revealing retrieval check

> Specify source IDs, versions, eligibility and the expected fallback.

Your evidence record

- Room-index practice: index v1 returns N1 assigning room A. Supplied corpus v2 supersedes N1 with N2 assigning room B. A stale returned record cannot establish current approval. Refresh against the authorised corpus or report unresolved freshness. Separate library case: approved current LIB-v2 says Wednesday closes16:00; older generated output says15:00. If the notice is unavailable, current time is unverified.

- Reference for this exercise: open this week’s student lesson, “Model Memory vs Retrieval”, at “See how it works” and “Follow a worked example”. Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Inspect a new student corpus result independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Evaluate the supplied search result for a student request under the stated public, teacher-only and withdrawn labels. No real documents or live service are provided. Explain the defect and where the system must enforce the relevant conditions.

Corpus has public P, teacher-only T and withdrawn W. A student search returns all three. Identify the filtering defect and whether hiding T after generation is sufficient.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

The Study Buddy retrieves course passages and never presents model memory as a citation.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for **Retrieval** and **index**ing. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Retrieval** and **index**ing. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A model states the library opens at 8 from learned patterns. A supplied current notice says 9. Explain the stronger answer and one remaining check.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

One assistant answers without search; another retrieves a handbook paragraph.

Compare **evidence** trails.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A model recalls last year's opening hours; a retrieved current notice gives different hours. How should the answer use this **evidence**?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 17 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A model states the library opens at 8 from learned patterns. A supplied current notice says 9. Explain the stronger answer and one remaining check.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Corpus has public P, teacher-only T and withdrawn W. A student search returns all three. Identify the filtering defect and whether hiding T after generation is sufficient.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

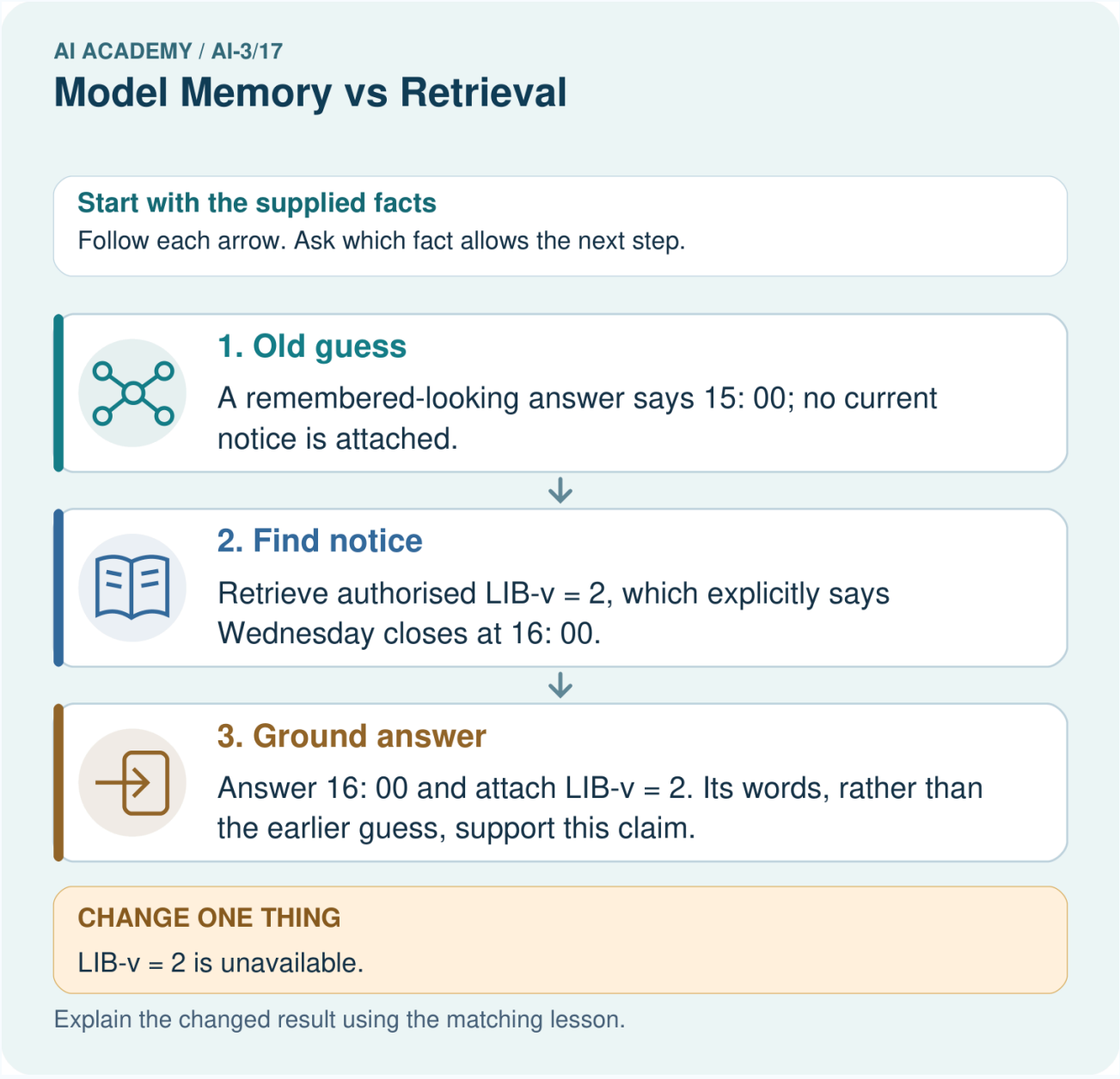
> Keep the explanation and the support you used.

Use this when helpful

The library assistant remembers an old opening time, but a current notice can be retrieved.

Approved notice LIB-v = 2 says Wednesday closes at 16: 00; the model previously produced 15: 00. The question asks the current Wednesday closing time.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

Why can the assistant not certify 15: 00 when the notice is missing? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- The library assistant remembers an old opening time, but a current notice can be retrieved.
- Which labelled step matches this detail? Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim.
- 1. Old guess

- 2. Find notice
- 3. Ground answer
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Ground answer Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. Read the picture in order.

  Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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> ---
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> ---
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> ---

▼ **Check the first explanation**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: LIB-v = 2 is unavailable. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.
- 3. Report that the current time cannot be verified.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: Report that the current time cannot be verified. The unsupported old guess does not become current **evidence** when **retrieval** fails.

> **My selected conclusion, deciding fact and explanation**
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> ---
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> ---
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> ---
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> ---

▼ **Check the changed case**

Repair a missing building block

Repair and save your work

Review Verify Numbers and Dates (ai-3/16). Goal: Verify a numeric claim using its source, denominator and units. Short practice: Which part can be corrected immediately from existing **evidence**, and which needs follow-up?. Return to this week and explain what you changed.

Review Hallucination Failure Types (ai-3/14). Goal: Explain the **mechanism** and terminology of generation failures and verification. Short practice: Can we conclude there will definitely be no lunch from S alone?.

Return to this week and explain what you changed.

Review Visual Composition (ai-3/11). Goal: Check object counts, spatial relations and arrow meaning separately. Short practice: Which checks pass after that single change, and what still needs repair?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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> | | |
> |---|---|
> | | |
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> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

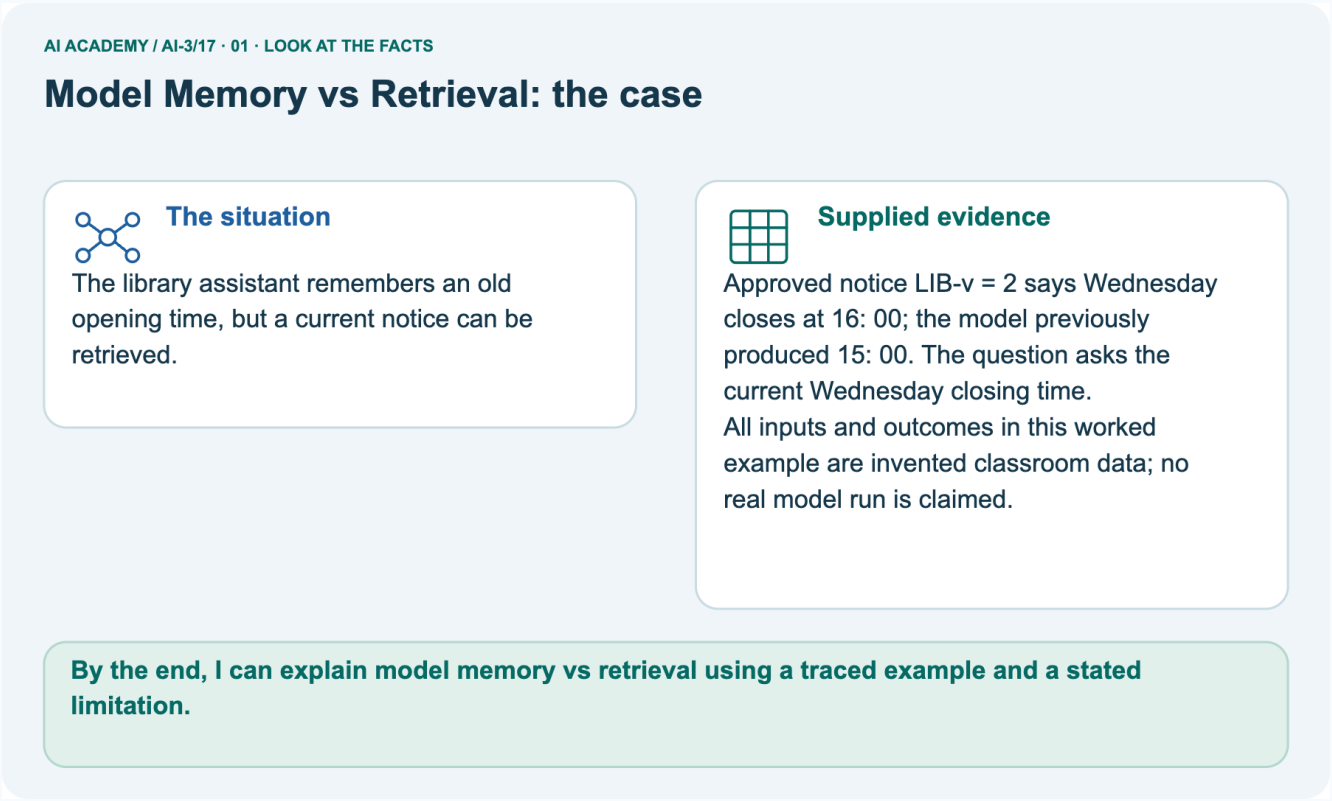
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Model Memory vs Retrieval: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

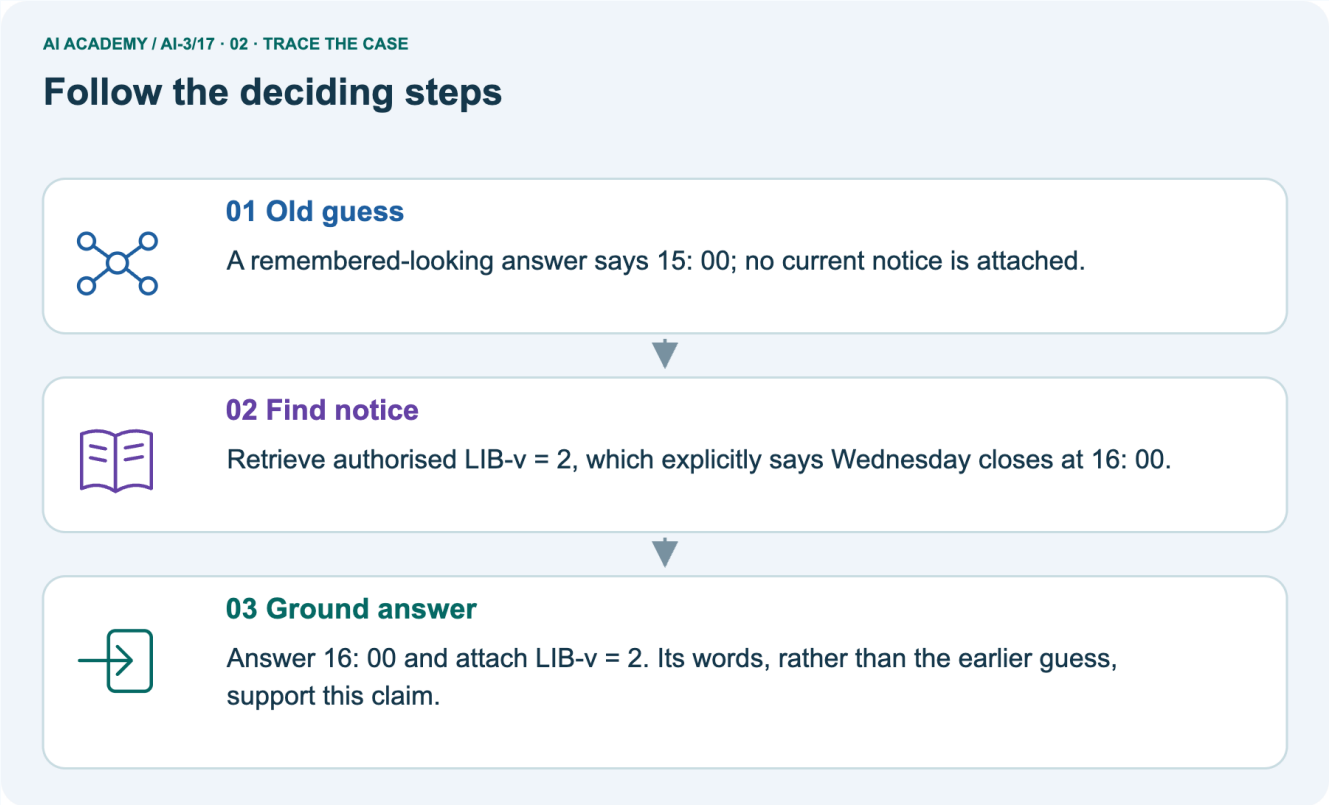
The library assistant remembers an old opening time, but a current notice can be retrieved. Supplied facts: Approved notice LIB-v = 2 says Wednesday closes at 16: 00; the model previously produced 15: 00.

The question asks the current Wednesday closing time. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

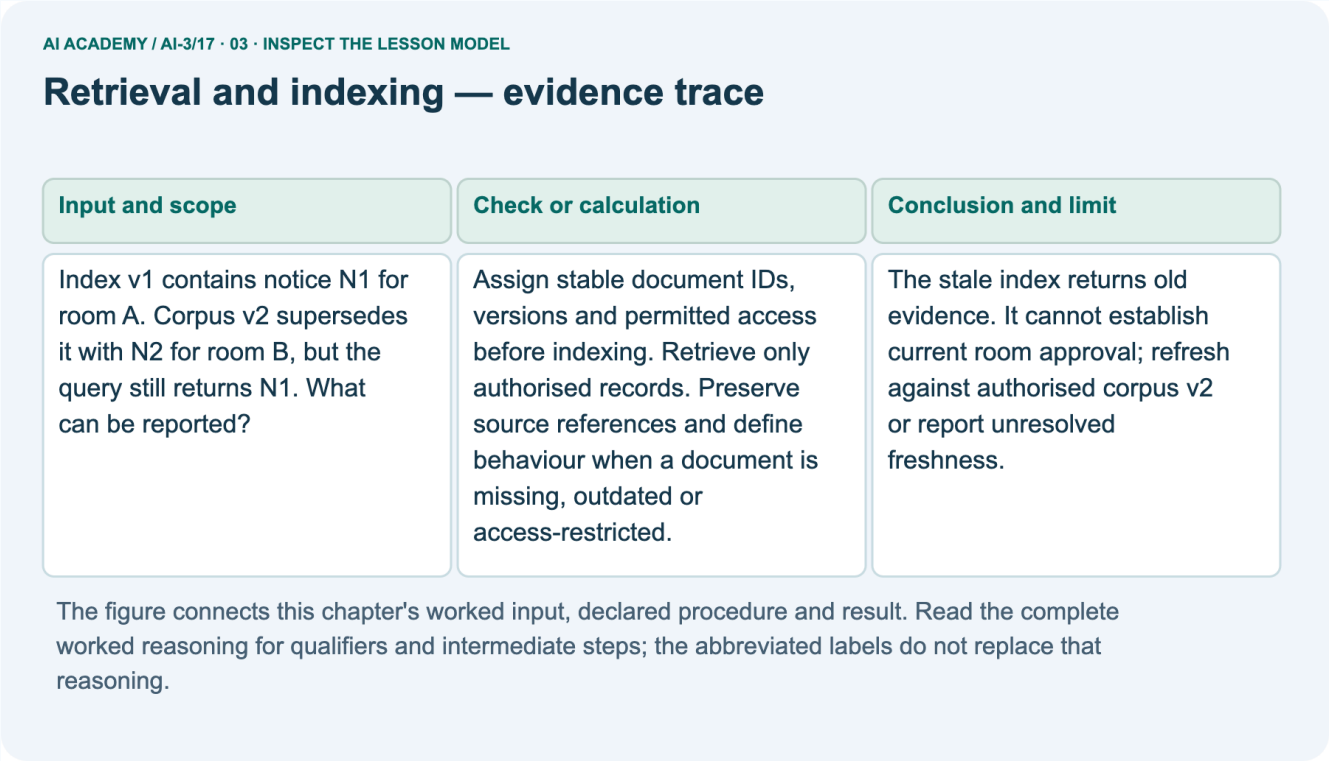
Old guess: A remembered-looking answer says 15: 00; no current notice is attached. → Find notice: Retrieve authorised LIB-v = 2, which explicitly says Wednesday closes at 16: 00. → Ground answer: Answer 16: 00 and attach LIB-v = 2.

Its words, rather than the earlier guess, support this claim.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Retrieval and indexing — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieval and **index**ing — **evidence** trace. **Index** v1 contains notice N1 for room A. **Corpus** v2 supersedes it with N2 for room B, but the query still returns N1.

What can be reported? | Assign stable document IDs, versions and permitted access before **index**ing. Retrieve only authorised records.

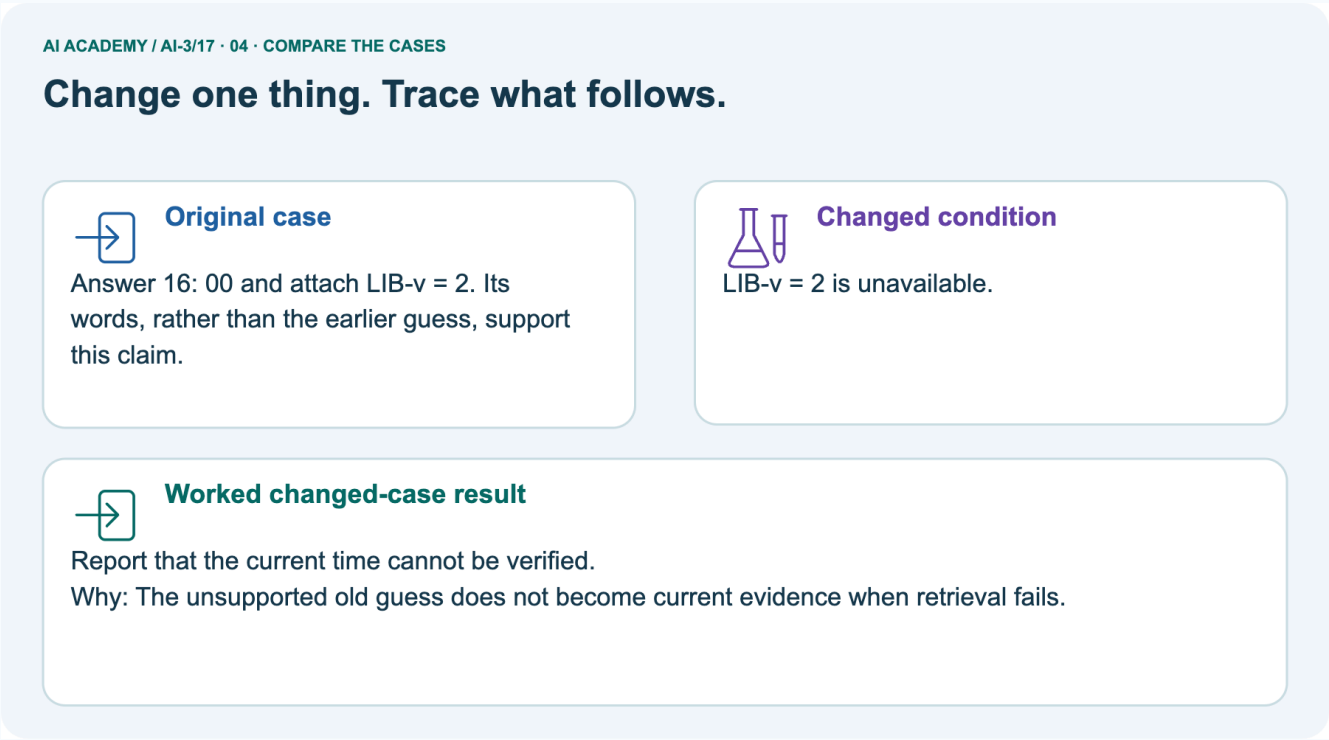
Preserve source references and define behaviour when a document is missing, outdated or access-restricted. | The stale **index** returns old **evidence**.

It cannot establish current room approval; refresh against authorised **corpus** v2 or report unresolved **freshness**.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

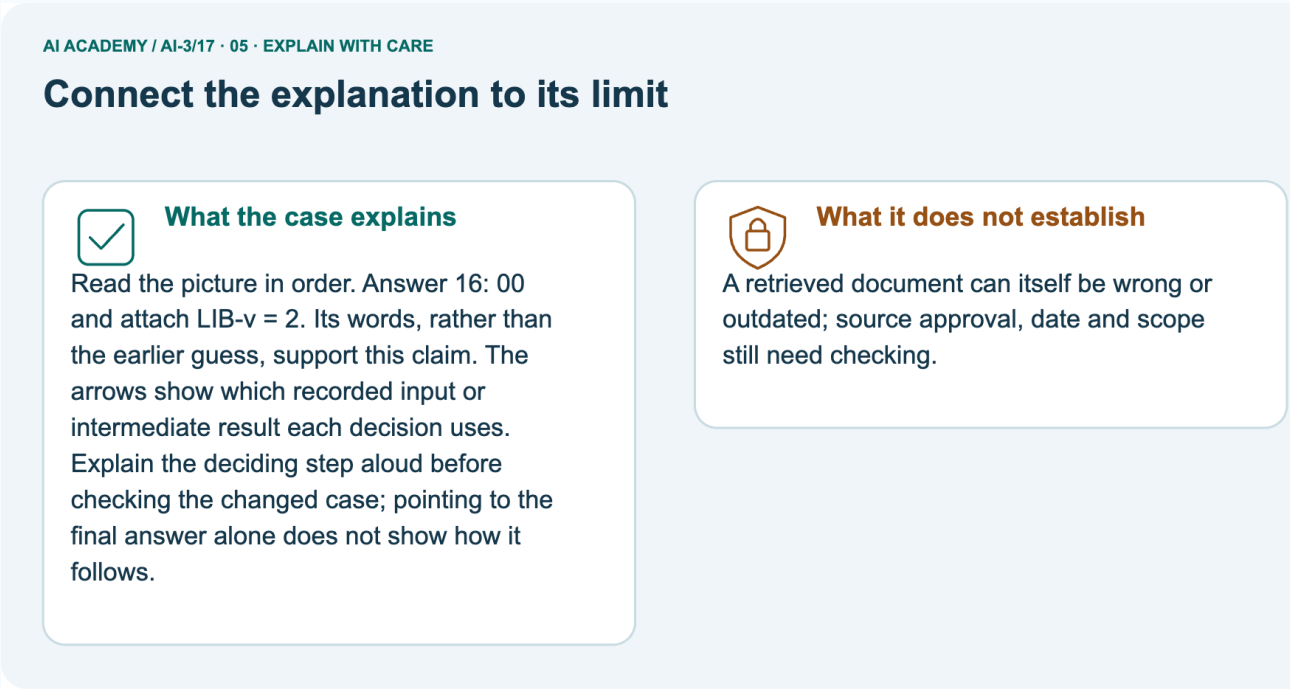
Original case: Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. Changed condition: LIB-v = 2 is unavailable. Expected result: Report that the current time cannot be verified.

Reason: The unsupported old guess does not become current **evidence** when **retrieval** fails.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Answer 16: 00 and attach LIB-v = 2. Its words, rather than the earlier guess, support this claim. The arrows show which recorded **input** or intermediate result each decision uses.

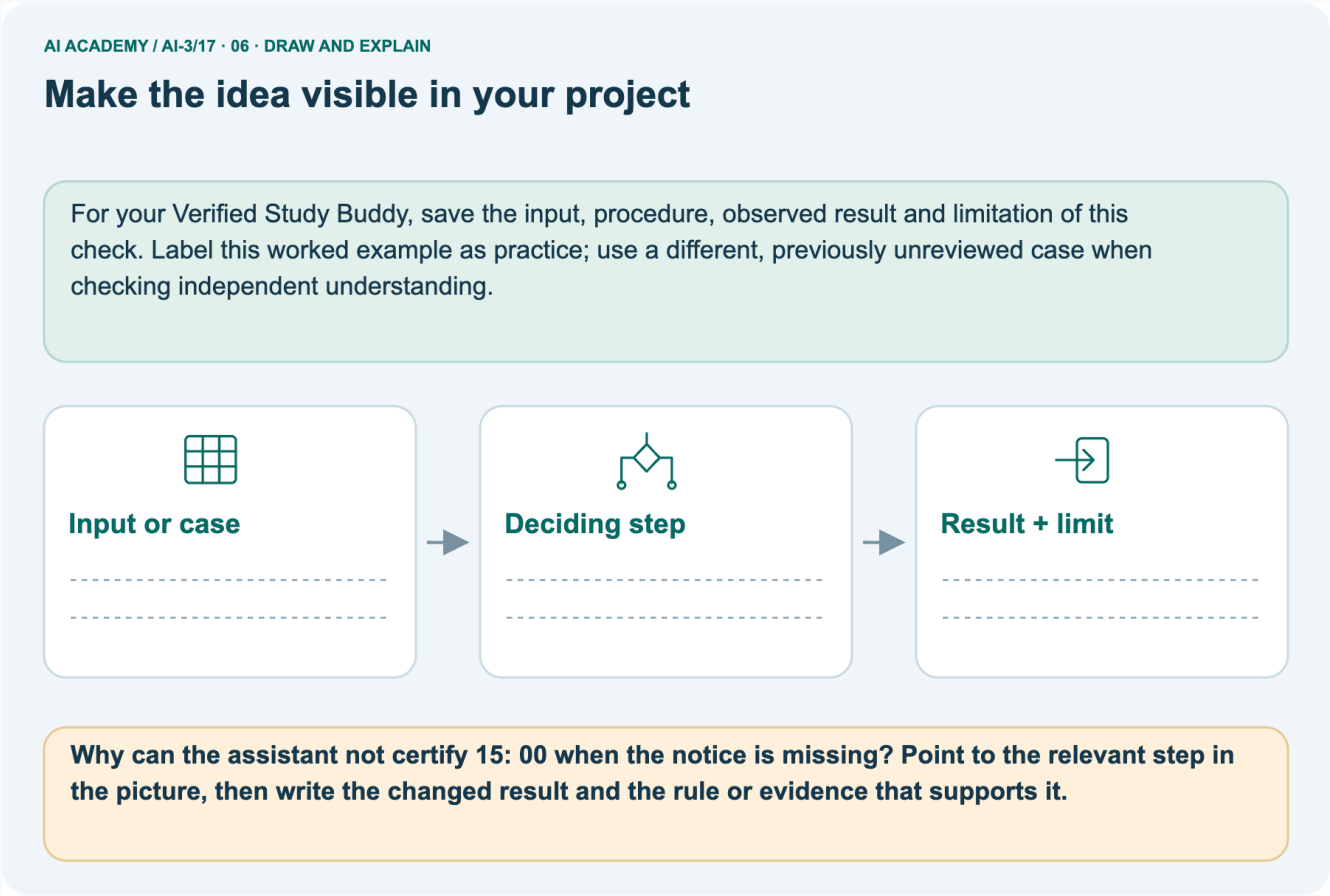
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A retrieved document can itself be wrong or outdated; source approval, date and scope still need checking.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can the assistant not certify 15: 00 when the notice is missing? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

> **My picture explanation, deciding evidence and one limit**

▼ **Earlier wording, alternatives and picture transcripts**

01. Model Memory vs Retrieval: the case

02. Follow the deciding steps

03. Retrieval and indexing — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.